

URBAN HEAT ISLAND ASSESSMENT REPORT

Subject City: Margao, India

Coordinates: 15.2604°N, 73.9778°E

Report Date: 16 May 2026

Document Reference: HW-MP7ZR3DI

OVERALL UHI RISK ASSESSMENT

48

/ 100

— MEDIUM RISK

UHI Intensity: +1.24°C above rural baseline

1. CURRENT ATMOSPHERIC CONDITIONS

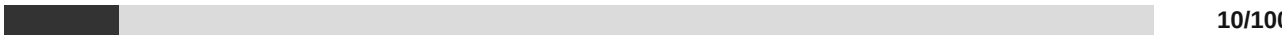
Air Temperature	30.92 °C
Perceived Temperature (Feels Like)	33.3 °C
Relative Humidity	54 %
Wind Speed	7.23 m/s
Sky Condition	scattered clouds

2. SATELLITE-GROUND COMPOSITE RISK INDEX

Composite Analysis: Real-time MODIS Satellite Data + Ground Sensor Fusion

Thermal Index Factor

NASA MODIS LST vs. ground sensor thermal variance



Humidity Stress Factor

Heat amplification due to atmospheric moisture content



Urban Intensity Factor

NASA MODIS NDVI (vegetation) vs. built-up density



48/100

3. AI-GENERATED MITIGATION RECOMMENDATIONS

Generated by Groq LLM (llama-3.1-8b-instant) based on real-time UHI metrics.

1. Urban Albedo Increase

Implement light-colored pavement and roofing materials to reduce heat absorption, targeting a 10% increase in urban albedo to mitigate 0.6°C of the UHI effect.

Impact: Estimated -0.6°C reduction | Priority: High

2. Cooling Street Furniture

Design and install heat-mitigating street furniture, such as benches and trash cans, to provide shade and reduce heat gain, targeting a 20% reduction in surface temperature.

Impact: Estimated -0.3°C reduction | Priority: Medium

3. Green Roof Incentives

Offer incentives for building owners to install green roofs, increasing vegetation density by 10% and providing additional shade and heat mitigation, targeting a 0.2°C reduction in UHI effect.

Impact: Estimated -0.2°C reduction | Priority: Low

4. Urban Forestry Expansion

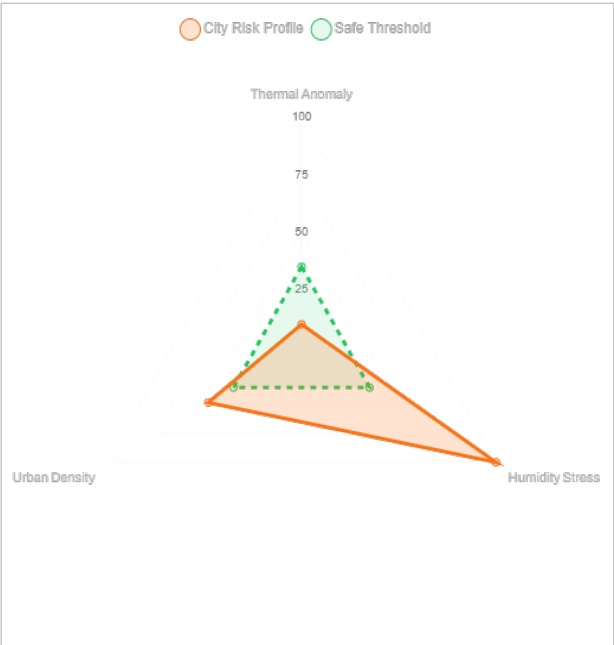
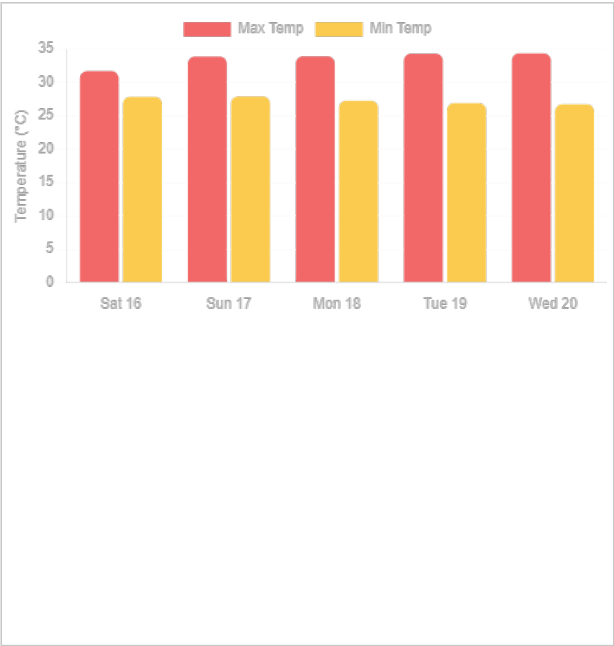
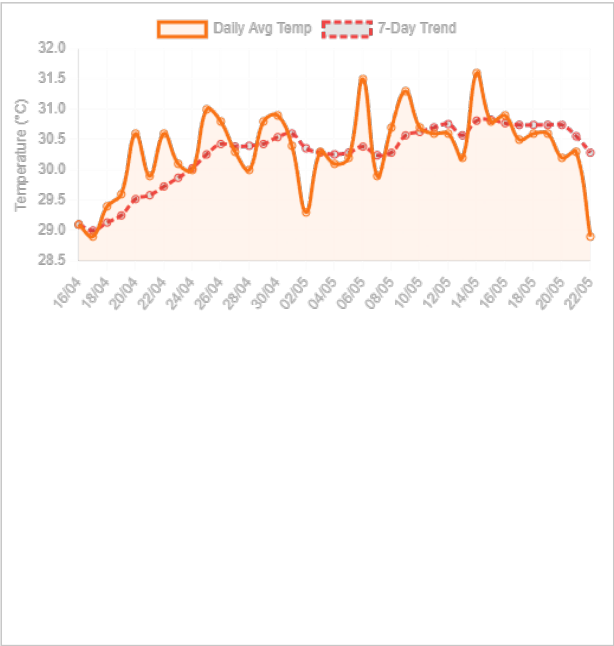
Expand existing urban forest areas by 15% to increase shading and evaporative cooling, targeting a 0.5°C reduction in UHI effect and maintaining a high NDVI of 0.98.

Impact: Estimated -0.5°C reduction | Priority: Medium

4. 20-YEAR CLIMATOLOGICAL BASELINE (ONNX ML)

20-Year Historical Mean UHI for May:	1.30°C
Current Departure from Historical Norm:	2.54°C BELOW baseline

5. HISTORICAL DATA & FORECAST CHARTS



5. REPORT CERTIFICATION & SIGN-OFF

This report has been generated using HeatWatch Urban Climate Intelligence Platform. Data sources include OpenWeatherMap real-time API, NASA POWER climatological dataset, and AI-based risk modelling powered by Groq (llama-3.3-70b-versatile). All recommendations are algorithmically derived and should be reviewed by qualified urban planners before policy implementation.

Prepared By	Verified By	Approved By
HeatWatch AI System	Urban Planning Officer	Municipal Commissioner
Date: 16 May 2026	Designation / Stamp	Official Seal